

- Blood dyscrasias, though still rare, may be more common in the elderly

Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Safety and efficacy have not been established

Pregnancy

- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients this may mean continuing treatment during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients particularly concerned about sexual side effects
- Patients with symptoms of anxiety
- As an augmenting agent to boost the efficacy of other antidepressants

- Patients with cardiovascular disease

Potential Disadvantages

- Patients particularly concerned about gaining weight
- Patients with low energy

Primary Target Symptoms

- Depressed mood
- Sleep disturbance
- Anxiety

Pearls

- * Adding alpha 2 antagonism to agents that block serotonin and/or norepinephrine reuptake may be synergistic for severe depression
- Adding mianserin to venlafaxine or SSRIs may reverse drug-induced anxiety and insomnia
- Efficacy of mianserin for depression in cancer patients has been shown in small controlled studies
- * Only causes sexual dysfunction infrequently
- Generally better tolerated than TCAs, including safer in overdose
- General lack of cardiovascular toxicity

Suggested Reading

Brogden RN , Heel RC , Speight TM , Avery GS . Mianserin: a review of its pharmacological properties and therapeutic efficacy in depressive illness . *Drugs* 1978 ;**16** (4):273 –301 .

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De Ridder JJ . Mianserin: result of a decade of antidepressant research . *Pharm Weekbl Sci* 1982 ;**4** (5):139–45 .

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Leinonen E , Koponen H , Lepola U . Serum mianserin and ageing . *Prog Neuropsychopharmacol Biol Psychiatry* 1994 ;**18** (5):833–45 .

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Rotzinger S , Bourin M , Akimoto Y , Coutts RT , Baker GB . Metabolism of some “second-” and “fourth-” generation antidepressants: iprindole, viloxazine, bupropion, mianserin, maprotiline, trazodone, nefazodone, and venlafaxine . *Cell Mol Neurobiol* 1999 ;**19** (4):427–42 .

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Wakeling A . Efficacy and side effects of mianserin, a tetracyclic antidepressant . *Postgrad Med J* 1983 ;**59** (690):229–31 .

Midazolam

Therapeutics

Brands

- Versed
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

Commonly Prescribed for

(bold for FDA approved)

- **Sedation in pediatric patients**
- **Sedation (adjunct to anesthesia)**
- **Preoperative anxiolytic**
- **Drug-induced amnesia**
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Intravenous injection: onset 3–5 minutes
- Intramuscular injection: onset 15 minutes, peak 30–60 minutes

If It Works

- Patients generally recover 2–6 hours after awakening

If It Doesn't Work

- Increase the dose
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmenting agents have not been adequately studied

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia

Notable Side Effects

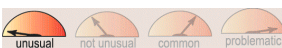
- Oversedation, impaired recall, agitation, involuntary movements, headache
- Nausea, vomiting
- Hiccups, fluctuation in vital signs, irritation/pain at site of injection
- Hypotension



Life-Threatening or Dangerous Side Effects

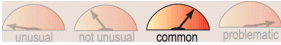
- Respiratory depression, apnea, respiratory arrest
- Cardiac arrest

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Intravenous (adults): 1–2.5 mg
- Liquid (age 16 and under): 0.25–1.0 mg/kg

Dosage Forms

- Syrup 2 mg/mL
- Nasal 5 mg/spray
- Intravenous 1 mg/mL, 5 mg/mL
- Intramuscular solution 50 mg/10 mL (5 mg/mL)

How to Dose

- Liquid single dose: 0.25–1.0 mg/kg; maximum dose generally 20 mg
- Intravenous (adults): administer over 2 minutes; monitor patient over the next 2 or more minutes to determine effects; allow 3–5 minutes between administrations; maximum 2.5 mg within 2 minutes



Dosing Tips

- Better to underdose, observe for effects, and then prudently raise dose while monitoring carefully

Overdose

- Sedation, confusion, poor coordination, respiratory depression, coma

Long-Term Use

- Not generally intended for long-term use

Habit Forming

- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- If administration was prolonged, do not stop abruptly

Pharmacokinetics

- Elimination half-life 1.8–6.4 hours
- Active metabolite



Drug Interactions

- If CNS depressants are used concomitantly, midazolam dose should be reduced by half or more
- Increased depressive effects when taken with other CNS depressants (see Warnings below)
- Drugs that inhibit CYP450 3A4, such as nefazodone and fluvoxamine, may reduce midazolam clearance and thus raise midazolam levels
- Midazolam decreases the minimum alveolar concentration of halothane needed for general anesthesia



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death

- Midazolam should be used only in an environment in which the patient can be closely monitored (e.g., hospital) because of the risk of respiratory depression and respiratory arrest
- Sedated pediatric patients should be monitored throughout the procedure
- Patients with COPD should receive lower doses
- Use with caution in patients with impaired respiratory function

Do Not Use

- If patient has angle-closure glaucoma
- If there is a proven allergy to midazolam or any benzodiazepine

Special Populations

Renal Impairment

- May have longer elimination half-life, prolonging time to recovery

Hepatic Impairment

- Longer elimination half-life; clearance is reduced

Cardiac Impairment

- Longer elimination half-life; clearance is reduced

Elderly

- Longer elimination half-life; clearance is reduced
- Intravenous: 1–3.5 mg; maximum 1.5 mg within 2 minutes

Children and Adolescents

- In most pediatric populations, pharmacokinetic properties are similar to those in adults
- Seriously ill neonates have reduced clearance and longer elimination half-life
- Hypotension has occurred in neonates given midazolam and fentanyl
- Intravenous dose: dependent on age, weight, route, procedure

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Midazolam crosses the placenta
- Neonatal flaccidity has been reported in infants whose mother took a benzodiazepine during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss
- Midazolam can be used to relieve postoperative pain after cesarean section

The Art of Psychopharmacology

Potential Advantages

- Fast onset
- Parenteral dosage forms

Potential Disadvantages

- Can be oversedating

Primary Target Symptoms

- Anxiety

Pearls

- Recovery (e.g., ability to stand/walk) generally takes from 2–6 hours after wakening
- Half-life may be longer in obese patients

- Patients with premenstrual syndrome may be less sensitive to midazolam than healthy women throughout the cycle
- Midazolam clearance may be reduced in postmenopausal women compared to premenopausal women
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Blumer JL . Clinical pharmacology of midazolam in infants and children .
Clin Pharmacokinet 1998 ;**35** :37 –47 .

Fountain NB , Adams RE . Midazolam treatment of acute and refractory status epilepticus . *Clin Neuropharmacol* 1999 ;**22** :261–7 .

Shafer A . Complications of sedation with midazolam in the intensive care unit and a comparison with other sedative regimens . *Crit Care Med* 1998 ;**26** :947–56 .

Yuan R , Flockhart DA , Balian JD . Pharmacokinetic and pharmacodynamic consequences of metabolism-based drug interactions with alprazolam, midazolam, and triazolam . *J Clin Pharmacol* 1999 ;**39** :1109 –25.

Milnacipran

Therapeutics

Brands

- Toledomin
- Ixel
- Savella
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine reuptake inhibitor (SN-RI)
- SNRI (dual serotonin and norepinephrine reuptake inhibitor); antidepressant; chronic pain treatment

Commonly Prescribed for

(bold for FDA approved)

- **Fibromyalgia**
- Major depressive disorder

- Neuropathic pain/chronic pain

How the Drug Works

- Boosts neurotransmitters serotonin, norepinephrine/noradrenaline, and dopamine
- Blocks serotonin reuptake pump (serotonin transporter), presumably increasing serotonergic neurotransmission
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Presumably desensitizes both serotonin 1A receptors and beta adrenergic receptors
- * Weak noncompetitive NMDA receptor antagonist (high doses), which may contribute to actions in chronic pain
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, milnacipran can increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms in depression

If It Works

- The goal of treatment of depression is complete remission of current symptoms as well as prevention of future relapses
- The goal of treatment of fibromyalgia and chronic neuropathic pain is to reduce symptoms as much as possible, especially in combination with other treatments
- Treatment of depression most often reduces or even eliminates symptoms, but is not a cure since symptoms can recur after medicine stopped
- Treatment of fibromyalgia and chronic neuropathic pain may reduce symptoms, but rarely eliminates them completely, and is not a cure since symptoms can recur after medicine is stopped
- Continue treatment of depression until all symptoms are gone (remission)
- Once symptoms of depression are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in fibromyalgia and chronic neuropathic pain may also need to be indefinite, but long-term treatment is not well studied in these conditions

If It Doesn't Work

- Many depressed patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other depressed patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some depressed patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation experience is limited compared to other antidepressants
- Benzodiazepines can reduce insomnia and anxiety
- Adding other agents to milnacipran for treating depression could follow the same practice for augmenting SSRIs or other SNRIs if done by experts while monitoring carefully in difficult cases

- Although no controlled studies and little clinical experience, adding other agents for treating fibromyalgia and chronic neuropathic pain could theoretically include gabapentin, tiagabine, other anticonvulsants, or even opiates if done by experts while monitoring carefully in difficult cases
- Mirtazapine, bupropion, reboxetine, atomoxetine (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, or treatment-resistant depression
- Hypnotics or trazodone for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- Check blood pressure before initiating treatment and regularly during treatment

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in serotonin and norepinephrine concentrations at receptors in parts of the brain and body other than

those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on acetylcholine release causing urinary retention or constipation)

- Most side effects are immediate but often go away with time

Notable Side Effects

- Most side effects increase with higher doses, at least transiently
- Headache, nervousness, insomnia, sedation
- Nausea, diarrhea, decreased appetite
- Sexual dysfunction (abnormal ejaculation/orgasm, impotence)
- Asthenia, sweating
- SIADH (syndrome of inappropriate antidiuretic hormone secretion)
- Dose-dependent increased blood pressure
- Dry mouth, constipation
- Dysuria, urological complaints, urinary hesitancy, urinary retention
- Increase in heart rate
- Palpitations



Life-Threatening or Dangerous Side Effects

- Rare induction of mania

- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)
- Rare seizures

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- In a few weeks, switch or add other drugs

Best Augmenting Agents for Side Effects

* For urinary hesitancy, give an alpha 1 blocker such as tamsulosin or naftopidil

- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Benzodiazepines for anxiety, agitation
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of milnacipran

Dosing and Use

Usual Dosage Range

- 30–200 mg/day in 2 doses

Dosage Forms

- Capsule 25 mg, 50 mg (France, other European countries, and worldwide markets)
- Capsule 15 mg, 25 mg, 50 mg (Japan)
- Tablet 12.5 mg, 25 mg, 50 mg, 100 mg

How to Dose

- Should be administered in 2 divided doses
- Initial 12.5 mg once daily; increase to 25 mg/day in 2 divided doses on day 2; increase to 50 mg/day in 2 divided doses on day 4; increase to 100 mg/day in 2 divided doses on day 7; maximum dose generally 200 mg/day



Dosing Tips

- Preferred dose for fibromyalgia may be 100 mg twice daily
- Higher doses usually well tolerated in fibromyalgia patients
- * Once daily dosing has far less consistent efficacy, so only give as twice daily
- Higher doses (>200 mg/day) not consistently effective in all studies of depression
- Nevertheless, some patients respond better to higher doses (200–300 mg/day) than to lower doses
- Different doses in different countries
- Different doses in different indications and different populations

- Preferred dose for depression may be 50 mg twice daily to 100 mg twice daily in France
 - Preferred dose for depression in the elderly may be 15 mg twice daily to 25 mg twice daily in Japan
 - Preferred dosing for depression in other adults may be 25 mg twice daily to 50 mg twice daily in Japan
- * Thus, clinicians must be aware that titration of twice daily dosing across a 10-fold range (30 mg–300 mg total daily dose) can optimize milnacipran's efficacy in broad clinical use
- Patients with agitation or anxiety may require slower titration to optimize tolerability
 - No pharmacokinetic drug interactions (not an inhibitor of CYP450 2D6 or 3A4)
 - As milnacipran is a more potent norepinephrine reuptake inhibitor than a serotonin reuptake inhibitor, some patients may require dosing at the higher end of the dosing range to obtain robust dual SNRI actions
 - At high doses, NMDA glutamate antagonist actions may be a factor

Overdose

- Vomiting, hypertension, sedation, tachycardia
- The emetic effect of high doses of milnacipran may reduce the risk of serious adverse effects

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper is prudent, but usually not necessary

Pharmacokinetics

- Half-life 8 hours
- No active metabolite



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing milnacipran
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)